

Solar PV Yield Assessment Report

Generated with PV Report Generator

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Germany
(48.1086°N, 11.6253°E)

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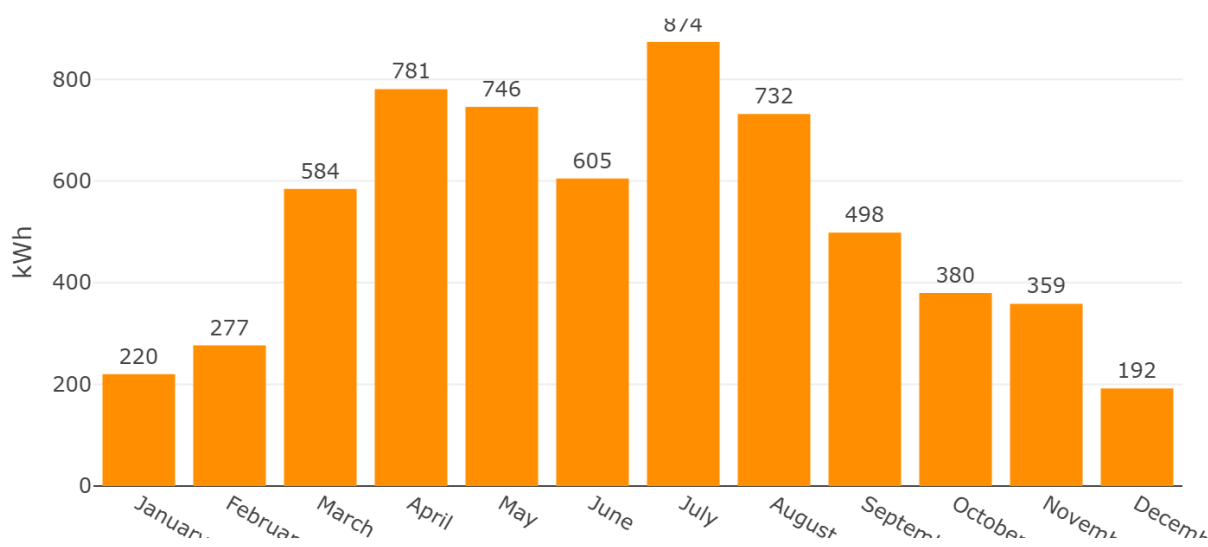
Executive Summary

Annual Yield (kWh)	6,247 kWh
Specific Yield (kWh/kWp)	1,249 kWh/kWp
Capacity Factor (%)	14.3%
Performance Ratio (%)	84.5%

Monthly Production

Month	Energy (kWh)
January	220.2
February	276.7
March	584.3
April	780.8
May	746.0
June	604.9
July	873.5
August	731.7
September	498.5
October	379.6
November	358.6
December	191.8
Total	6,246.8

Monthly Energy Production (kWh)



System Configuration

Latitude	48.1086°
Longitude	11.6253°
System Size (kWp)	5.0 kWp
Tilt Angle (°)	30.0°
Azimuth (°, 180=South)	180.0°
Inverter Efficiency (%)	96.0%
DC/AC Ratio	1.2
Ground Albedo	0.2
Transposition Model	Perez
Module Technology	Thin Film

Methodology

This report was generated using pvlib, an open-source Python library for photovoltaic energy modeling. Weather data is sourced from PVGIS (Photovoltaic Geographical Information System) Typical Meteorological Year (TMY) dataset provided by the European Commission Joint Research Centre.

The simulation pipeline includes:

- Solar position calculation (NREL SPA algorithm)
- Plane-of-array irradiance transposition (Perez model by default)
- Cell temperature estimation (SAPM model)
- DC power output (PVWatts model)
- AC conversion via constant inverter efficiency or PVWatts inverter model
- Monthly and annual energy yield aggregation

Assumptions & Limitations

- Weather data represents a Typical Meteorological Year and may not reflect actual conditions in any specific year.
- System losses are applied as a single derating factor unless detailed losses are specified in Pro Mode.
- No near-shading analysis is performed.
- Module degradation over time is not modeled unless explicitly set.
- Grid availability and curtailment are not considered.
- Results are estimates and should be validated with on-site measurements for financial decisions.